

Understanding Analysis, 2nd Edition

Chapter 6 - Section 6.2 - Uniform Convergence of a Sequence of Functions

1. Let

$$f_n(x) = \frac{nx}{1 + nx^2}$$

- (a) Find the pointwise limit of (f_n) for all $x \in (0, \infty)$.
- (b) Is the convergence uniform on $(0, \infty)$?
- (c) Is the convergence uniform on $(0, 1)$?
- (d) Is the convergence uniform on $(1, \infty)$?

Solution:

- (a) Intuitively, as n grows the $+1$ term starts to not matter, so it should converge to $\frac{nx}{nx^2} = \frac{1}{x}$. Formally

$$\lim_{n \rightarrow \infty} \frac{nx}{1 + nx^2} = \lim_{n \rightarrow \infty} \frac{x}{1/n + x^2} = 1/x$$

- (b) No. Given $\epsilon > 0$, let N such that for all $n \geq N$, $|f_n(x) - \frac{1}{x}| < \epsilon$. Or in other words

$$\left| \frac{nx}{1 + nx^2} - \frac{1}{x} \right| = \frac{1}{x(1 + nx^2)} < \epsilon$$

For this to be uniformly convergent, the N needs to work for all x . But the lhs is unbounded as $x \rightarrow 0$; there's always a small enough x such that the lhs is eventually greater than ϵ .

- (c) No, for the same reason as part (b). We can always choose small enough x to break the uniform convergence.
- (d) Now the answer is yes. Reusing the result from part (b), for $x > 1$, the value is upper-bounded by $x = 1$

$$\frac{1}{x(1 + nx^2)} < \frac{1}{1 + n}$$

This doesn't depend on x , so we can simply choose $N \geq \frac{1}{\epsilon} - 1$.

2. (a) Define a sequence of functions on $\mathbf{R}$ by

$$f_n(x) = \begin{cases} 1 & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

and let f be the pointwise limit of f_n .

Is each f_n continuous at zero? Does $f_n \rightarrow f$ uniformly on $\mathbf{R}$? Is f continuous at zero?

- (b) Repeat this exercise using the sequence of functions

$$g_n(x) = \begin{cases} x & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

- (c) Repeat once more with

$$h_n(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \\ x & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-1} \\ 0 & \text{otherwise} \end{cases}$$

In each case, explain how the results are consistent with the Continuous Limit Theorem (Theorem 6.2.6).

Solution:

- (a) Yes, each f_n is continuous at zero. Simply choose $\delta < \frac{1}{n}$, then $f(x) = 0$ for $x < \delta$.

To answer uniform convergence, let's first define f . For x not of the form $\frac{1}{n}$, $f(x)$ is clearly 0. For x of the form $\frac{1}{n}$, then for all $m \geq n$, $f_m(x) = 1$ i.e. eventually all $f_n(x)$ becomes 1. So f is

$$f(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

It's not uniformly convergent. Let $\epsilon < 1$. Now say we choose $N = 5$, then for $x = \frac{1}{6}$, we have $f_5(\frac{1}{6}) = 0$ but $f(\frac{1}{6}) = 1$. In general, no single N can work for all x because we can choose $x = \frac{1}{N+1}$, then $|f_N(x) - f(x)| = 1 > \epsilon$.

Finally, f is not continuous at zero. We have $f(0) = 0$. But since $\frac{1}{n} \rightarrow 0$, there's always x of the form $\frac{1}{n}$ close enough to zero such that $f(x) = 1$.

- (b) Each g_n is continuous at zero. Same argument as part (a), choose $\delta < \frac{1}{n}$ then $|x| < \delta \implies g_n(x) = 0$.

Using a very similar argument as part (a), the limit function g is

$$g(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

But unlike part(a), this is uniformly convergent. The idea is to choose N big enough such that even if $g_N(x) = 0$, then x needs to be really small, hence $|g_N(x) - g(x)| = |0 - x| = |x|$ is really small as well.

Formally, choose $N > \frac{1}{\epsilon}$. Then for $n \geq N$, either $g_n(x) = g(x)$ or if $g_n(x) = 0$, then $x = \frac{1}{m}$ with $m > n$. But since $n > \frac{1}{\epsilon}$, if $m > n$ then $\frac{1}{m} < \epsilon$, hence $|g_n(\frac{1}{m}) - g(\frac{1}{m})| = \frac{1}{m} < \epsilon$.

g is continuous at zero, as $\frac{1}{n}$ converges to zero.

- (c) Each h_n is continuous at zero. Choose $\delta < \frac{1}{n}$, then $|x| < \delta \implies h_n(x) = 0$.

Using a similar argument as part (a), for each x of the form $\frac{1}{n}$, we have $h_m(x) = x = \frac{1}{n}$ for all $m > n$. So h is

$$h(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

h is the same as g from part(b), but in this case it's not uniformly convergent. The issue here is that

$$h_N(\frac{1}{N}) - h(\frac{1}{N}) = 1 - \frac{1}{N}$$

So as $x = \frac{1}{n}$ gets smaller, we need to choose increasingly larger N .

More formally, let $\epsilon < \frac{1}{2}$ and $x = \frac{1}{2}$. Choose $N \geq 3$. Then for $n \geq N$, we have

$$h_n(\frac{1}{2}) - h(\frac{1}{2}) = \frac{1}{2} - \frac{1}{2} = 0$$

But this same N cannot for $x = \frac{1}{N+1}$ because for $n = N + 1$ we have

$$h_{N+1}(\frac{1}{N+1}) - h(\frac{1}{N+1}) = 1 - \frac{1}{N+1}$$

And since $N \geq 3$, then $\frac{1}{N+1} \leq \frac{1}{4}$, hence

$$1 - \frac{1}{N+1} \geq 1 - \frac{1}{4} > \epsilon$$

Finally, h is continuous at zero as it's the same as g in part(b).

3. For each $n \in \mathbf{N}$ and $x \in [0, \infty)$, let

$$g_n(x) = \frac{x}{1+x^n}$$

and

$$h_n(x) = \begin{cases} 1 & \text{if } x \geq 1/n \\ nx & \text{if } 0 \leq x < 1/n \end{cases}$$

- (a) Find the pointwise limit on $[0, \infty)$.
- (b) Explain how we know that the convergence *cannot* be uniform on $[0, \infty)$.
- (c) Choose a smaller set over which the convergence is uniform and supply a proof.

Solution:

(a)

$$g(x) = \begin{cases} x & \text{if } x < 1 \\ 1/2 & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$

For h , note that for $x > 0$ and large enough n then $x \geq 1/n$ eventually and we always enter the first case.

For $x = 0$, we always enter the second case. So h is

$$h(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$

(b) For h_n , let's try $\epsilon = 1/4$ and $x = 1/2$ and see what we get.

$h_1(1/2) = \frac{1}{2}$ and $|h_1(1/2) - h(1/2)| = |1/2 - 1| = 1/2 > \epsilon$, so $N = 1$ doesn't work.

$h_2(1/2) = 1$. So $N \geq 2$ works because $h_N(1/2) = 1$.

But this N doesn't work for $x = \frac{1}{N^2}$, because $h_N(\frac{1}{N^2}) = \frac{1}{N}$ and even if we choose the minimum $N = 2$, we get $|\frac{1}{2} - 1| = \frac{1}{2} > \epsilon$. Choosing larger N will only increase the distance from 1 even more.

– **Better solution by chatgpt** –

Use the Continuous Limit Theorem. g_n and h_n is continuous for all n , so if they're uniformly convergent then g and h must be continuous. But g is discontinuous at $x = 1$ and h at $x = 0$.

- (c) For g_n , a simple option is to restrict to $x = 1$. Then $g_n(1) = g(1) = 1/2$ for all n which is certainly uniformly convergent. Likewise for h_n , restrict to $x = 0$. Then $h_n(0) = h(0) = 0$ for all n . But such trivial sets are probably not what the author intended. Let's choose a bigger set.

– **Incorrect** –

For g_n , we can stay away from the discontinuity at 1. Say, $(1, \infty)$. Then $g(x) = 0$ and

$$|g_n(x) - g(x)| = \frac{x}{1+x^n} \leq \frac{x}{x^n} = \frac{1}{x^{n-1}}$$

– **Corrected** –

Oops, $(1, \infty)$ doesn't work because x can still get arbitrarily close to 1. For any N , we can choose x close enough to 1 such that $\frac{1}{x^{n-1}}$ gets close to 1 hence far from $g(x) = 0$.

Instead, we need an interval bounded away from 1, say $[2, \infty)$. then

$$|g_n(x) - g(x)| \leq \frac{1}{x^{n-1}} \leq \frac{1}{2^{n-1}} \rightarrow 0$$

So for any ϵ , choose N big enough so that $\frac{1}{2^{N-1}} < \epsilon$.

h_n is uniformly convergent on $[1, \infty)$ because for $x \geq 1$ we'll always enter the first case, hence $h_n(x) = 1$ for all n and x .

4. TODO. It depends on Exercise 5.2.8 which I haven't done yet.
5. Using the Cauchy Criterion for convergent sequences of real numbers (Theorem 2.6.4), supply a proof for Theorem 6.2.5. (First, define a candidate for $f(x)$, and then argue that $f_n \rightarrow f$ uniformly).

Solution: The proof strategy is very similar to Theorem 2.6.4.

Necessity: uniform convergence $\implies$ Cauchy

There exists N such that for all $n \geq N$, $|f_n(x) - f(x)| < \epsilon/2$ for all x . Then for any $m, n \geq N$ and any x we have

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \epsilon/2 + \epsilon/2 = \epsilon$$

Satisfying Cauchy criterion.

Sufficiency: Cauchy $\implies$ uniform convergence

The Cauchy criterion doesn't say what the limit function is, so we need to define it

first. For a **particular** x , the sequence of real numbers $f_1(x), f_2(x), \dots$ is a Cauchy sequence, so it converges to a limit L_x . Define f to be

$$f(x) = L_x$$

Now we prove uniform convergence. By the Cauchy hypothesis (for sequence of functions), there exists N such that for all $n, m \geq N$, we have

$$|f_n(x) - f_m(x)| < \epsilon/2$$

for any x .

Also, by the fact that $f_1(x), f_2(x), \dots \rightarrow L_x$, there exists a $k \geq N$ such that $|f_k(x) - L_x| < \epsilon/2$. The value of k depends on x , but that's okay because for any x , $k \geq N$ and N does not depend on x .

Combining these two inequalities, we satisfy the condition for uniform convergence. For any $n \geq N$, we have

$$|f_n(x) - L_x| \leq |f_n(x) - f_k(x)| + |f_k(x) - L_x| < \epsilon/2 + \epsilon/2 = \epsilon$$

for any x .

6. TODO

7. Let f be uniformly continuous on all of $\mathbf{R}$, and define a sequence of functions by $f_n(x) = f(x + \frac{1}{n})$. Show that $f_n \rightarrow f$ uniformly. Give an example to show that this proposition fails if f is only assumed to be continuous and not uniformly continuous on $\mathbf{R}$.

Solution: We need to show there exists N such that for all $n \geq N$ and for all x

$$\left| f\left(x + \frac{1}{n}\right) - f(x) \right| < \epsilon$$

By uniform continuity, there exists $\delta > 0$ such that for all x , if $x - \delta < c < x + \delta$ then $|f(c) - f(x)| < \epsilon$. The idea is we just have to pick N (for the uniform convergence) large enough such that $\frac{1}{N} < \delta$.

Since $(\frac{1}{n}) \rightarrow 0$, choose N large enough such that $\frac{1}{N} < \delta$. Then for all $n \geq N$

$$x - \delta < x + \frac{1}{n} < x + \delta \implies \left| f\left(x + \frac{1}{n}\right) - f(x) \right| < \epsilon$$

For the counterexample, take $f(x) = x^2$. This is continuous but not uniformly continuous on $\mathbf{R}$. Then

$$\left| f\left(x + \frac{1}{n}\right) - f(x) \right| = \left| \frac{1}{n^2} + \frac{2x}{n} \right|$$

Suppose there's an N that works for a particular x_1 and ϵ . There exists another x_2 large enough such that $\frac{2x}{n}$ is eventually larger than ϵ , so the convergence is not uniform. More formally, let $\epsilon = 1$. If we choose $x = N$ then

$$\left| \frac{1}{n^2} + \frac{2x}{n} \right| = 2 + \frac{1}{n^2} > 1$$

As to how I knew that x^2 would work as a counterexample, I don't. I just chose the simplest non uniformly continuous function I can think of and it luckily worked. Chatgpt told me the intuition behind this is that " x^2 changes faster if x is large, so shifting by $1/n$ may be tiny, but at large x the vertical change can still be large".

8. Let (g_n) be a sequence of continuous functions that converges uniformly to g on a compact set K . If $g(x) \neq 0$ on K , show $(1/g_n)$ converges uniformly on K to $1/g$.

Solution: I missed the fact that we can use compactness to bound $g(K)$ and only realized it with chatgpt's assistance.

Given $\epsilon > 0$, we need to show that there exists N such that for all $n \geq N$ and for all $x \in K$

$$\left| \frac{1}{g_n(x)} - \frac{1}{g(x)} \right| = \frac{|g_n(x) - g(x)|}{|g_n(x)||g(x)|} < \epsilon$$

The difficulty lies in bounding the denominator. We use the fact that g is continuous on a compact set. By Theorem 4.4.1, $g(K)$ is compact, so it's bounded. And since $0 \notin g(K)$, then there exists a positive lower bound m such that $|g(x)| \geq m$, then

$$\frac{|g_n(x) - g(x)|}{|g_n(x)||g(x)|} \leq \frac{|g_n(x) - g(x)|}{|g_n(x)|m}$$

Since (g_n) converges uniformly to g , we can choose N_1 such that $g_n(x)$ is close to $g(x)$ or equivalently, $|g_n(x)|$ close to $|g(x)|$. Formally, this follows from the fact that

$$||g_n(x)| - |g(x)|| \leq |g_n(x) - g(x)|$$

Choose N_1 such that $|g_n(x)| \geq \frac{m}{2}$, resulting in

$$\frac{|g_n(x) - g(x)|}{|g_n(x)|m} \leq \frac{2}{m^2}|g_n(x) - g(x)|$$

Finally, choose N_2 such that $|g_n(x) - g(x)| < \frac{\epsilon m^2}{2}$ and choose $N = \max(N_1, N_2)$

$$\frac{2}{m^2}|g_n(x) - g(x)| < \frac{2}{m^2} \frac{\epsilon m^2}{2} = \epsilon$$

9. Assume (f_n) and (g_n) are uniformly convergent sequences of functions.

- (a) Show that $(f_n + g_n)$ is a uniformly convergent sequence of functions.
- (b) Give an example to show that the product $(f_n g_n)$ may not converge uniformly.
- (c) Prove that if there exists an $M > 0$ such that $|f_n| \leq M$ and $|g_n| \leq M$ for all $n \in \mathbf{N}$, then $(f_n g_n)$ does converge uniformly.

Solution:

- (a) Let $(f_n) \rightarrow f$ and $(g_n) \rightarrow g$ uniformly, then $(f_n + g_n) \rightarrow (f + g)$.

Choose N_f such that $|f_n(x) - f(x)| < \epsilon/2$ for all x . Choose N_g similarly, and let $N = \max(N_f, N_g)$. Then

$$|(f_n(x) + g_n(x)) - (f(x) + g(x))| \leq |f_n(x) - f(x)| + |g_n(x) - g(x)| < \epsilon/2 + \epsilon/2 = \epsilon$$

- (b) Let $f(x) = x$ and $g(x) = \frac{1}{n}$. $f_n \rightarrow f(x) = x$ uniformly, while $g_n \rightarrow g(x) = 0$ uniformly. But $(f_n g_n) = \frac{x}{n}$ does not converge uniformly to $(fg) = 0$ because there exists large enough x so that $\frac{x}{n}$ is big.

Concretely, let $\epsilon = 1/2$. For an N that works for a particular x_1 , the same N does not work for $x_2 = N$ because $\frac{N}{N} = 1 > \epsilon$.

- (c) First, note that if $|f_n| \leq M$ then $|f| \leq M$ as well by Squeeze (if $|f| > M$, then we can make $|f_n(x)|$ within $M + (|f| - M)/2$ of f , a contradiction). The rest follows the typical proof for proving convergence of products

$$\begin{aligned} |f_n(x)g_n(x) - f(x)g(x)| &= |f_n(x)g_n(x) - f(x)g_n(x) + f(x)g_n(x) - f(x)g(x)| \\ &\leq |g_n(x)||f_n(x) - f(x)| + |f(x)||g_n(x) - g(x)| \\ &\leq M|f_n(x) - f(x)| + M|g_n(x) - g(x)| \end{aligned}$$

Then choose N such that $|f_n(x) - f(x)|$ and $|g_n(x) - g(x)|$ are both less than $\frac{\epsilon}{2M}$.